

# A Branch-Oriented Architecture for Mathematical Foundations

A Methodology for Constructing, Documenting, and Comparing  
Formal Theories

[Author Name]

[Institution]

[email]

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## Abstract

Existing foundational systems for mathematics — axiomatic set theory, formal arithmetic, dependent type theories, and categorical foundations — are logically precise, expressively powerful, and internally consistent. What they do not provide, by design, is a structural record of their own construction: the genuine choices made at critical junctures of their development, the alternatives that were available but not taken, and the specific assumptions that represent new foundational commitments rather than derived consequences of prior content. This absence is not a logical deficiency but an architectural one.

We propose the *Branch-Oriented Mathematical Architecture* (BOMA), a framework for the organization, documentation, and comparison of mathematical foundations. BOMA introduces seven structural components: Blocks as the atomic units of foundational content; a dependency relation partially ordering Blocks within a development; Decision Points as explicit records of genuine foundational choices; Branches as independent developments originating from unchosen alternatives at Decision Points; Local Consistency as the sole criterion of branch validity, assessed independently of all other branches; Minimal Injection as the discipline governing the introduction of new assumptions; and Architectural Transparency as the conjunctive structural property produced by the full application of these requirements.

A central consequence of the framework is the Foundational Space: the collection of all locally consistent architectural paths sharing a common origin, equipped with a natural ultra-pseudometric structure. Within this space, contradictions between branches are not architectural failures but structural

information — precise measurements of the non-triviality of the Decision Points at which diverging branches separate. Comparison between branches is established as an epistemological instrument, not a validation procedure.

Every existing foundational system is shown to correspond to a locally consistent, architecturally opaque path within an appropriate Foundational Space. BOMA is a meta-framework: it makes no specific mathematical commitments and competes with no existing foundation. It adds a structural dimension that the existing paradigm does not supply, and constitutes a possibility space within which unknown mathematics occupies a determinate, reachable, and structurally characterized position.

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# 1 Introduction

## 1.1 The Success and the Silence

The foundational systems of modern mathematics are among the most carefully constructed intellectual artifacts in the history of human thought. Zermelo-Fraenkel set theory with the Axiom of Choice [20, 44], the Peano axioms in their first-order formulation [34], Martin-Löf Type Theory [26, 27], Homotopy Type Theory [41], and the categorical foundations developed from Lawvere’s work [23, 25] — each of these systems is internally precise to a degree that admits mechanical verification, expressively powerful enough to formalize the vast majority of contemporary mathematical practice, and consistent by all available evidence.

This success deserves acknowledgment before anything else is said. The architecture proposed in this paper is addressed to researchers who work within, or in close proximity to, this foundational tradition. Its starting point is not a failure of existing foundations but an observation about what those foundations, by the nature of their design, do not provide.

The observation is this: *existing foundational systems are silent about their own construction.*

A mathematician who opens a standard presentation of ZFC encounters nine axioms, stated in the language of first-order logic with a single primitive relation. The axioms are clearly labeled. The inference rules are declared. The derivations are checkable. Nothing in this presentation indicates that the adoption of the Axiom of Choice was contested, that the Axiom of Foundation was introduced to exclude pathological sets, that the first-order formulation was selected over a second-order alternative, or that the underlying logic could have been chosen to be intuitionistic rather than classical. These choices were made. Their consequences permeate every theorem derivable within ZFC. But the presentation of ZFC does not record them.

## 1.2 What Is Not Visible

The silence identified in §1.1 has three specific manifestations.

**The invisibility of Decision Points.** Every foundational system passed through moments of genuine choice. These moments — which we term *Decision Points* — are real architectural events. They determined the character of the mathematics that followed. But in the formal presentations of existing systems, they leave no trace.

**The suppression of alternatives.** A consequence of invisible Decision Points is that the space of foundational alternatives surrounding any existing system is not visible from within that system’s formal presentation.

**The conflation of injections and derivations.** Existing foundational presen-

tations do not systematically distinguish between statements that are genuine new assumptions and statements that are derived consequences of earlier assumptions.

*Remark 1.1* (Prior Work on Theory Structuring). Several existing research programs address aspects of the problem identified above. Theory structuring languages — most prominently CASL [36] and its predecessors — provide formal mechanisms for building theories from smaller components. The OMDOC format and the MMT framework [21, 35] provide comprehensive approaches to the representation of mathematical knowledge including theory graphs. Institution theory [15] provides a categorical framework for logical systems achieving a form of indifference to the specific logic employed.

The present paper does not claim that these frameworks are superseded by BOMA. Each addresses a genuine problem in the organization of formal mathematical knowledge. BOMA addresses a problem that none of them addresses: the explicit, structurally mandatory documentation of foundational decisions within the formal record of a mathematical development, together with the preservation of unchosen alternatives as first-class architectural objects.

### 1.3 The Architectural Question

The three absences described above suggest a question that is not, in the first instance, a mathematical question:

*Is the linear presentation of foundational mathematics the only organizational paradigm consistent with mathematical rigor? Or is there an alternative paradigm that preserves all the logical properties of the existing approach while adding a structural dimension that the existing approach does not supply?*

This is an architectural question. The present paper proposes such a framework: the Branch-Oriented Mathematical Architecture, hereafter **BOMA**.

The answer to this question has consequences beyond the organization of existing mathematics. A framework that makes foundational construction transparent and branchable is simultaneously a framework for mathematical exploration: every documented Decision Point is an entrance to mathematical territory that may not yet have been traversed.

### 1.4 Our Contribution

This paper makes the following contributions.

**A formal definition of Architectural Transparency.** We introduce a precise characterization of what it means for a foundational development to be fully legible:

to expose its dependency structure, its Decision Points, its injection structure, and its Branch Tree in a form computable from the structural record without reconstruction.

**A structural framework for foundational development.** We introduce seven structural components — Blocks, Dependencies, Decision Points, Branches, Local Consistency, Minimal Injection, and Architectural Transparency — and develop their formal properties in §3.

**The principle of Local Consistency as the sole criterion of branch validity.** We establish in §3.5 that a branch is valid if and only if it is internally consistent, independently of all other branches.

**The Foundational Space as a mathematical object.** We introduce in §4 the Foundational Space  $\mathfrak{F}(\beta_0)$ : the collection of all locally consistent architectural paths sharing a common Atomic Block, equipped with a natural ultra-pseudometric structure.

**A precise account of comparison as an epistemological tool.** We establish in §5 that structural comparison between branches is not a validation procedure but an epistemological instrument.

**The situation of existing foundations within BOMA.** We show in §6 that every existing foundational system corresponds to a locally consistent, architecturally opaque path in an appropriate Foundational Space.

A three-level logical architecture — object logic, framework logic, and meta-logic — is introduced in Remark 3.35, establishing the precise scope of the framework’s logical neutrality claims.

## 1.5 What This Paper Does Not Claim

This paper does not claim that existing foundations are incorrect, incomplete, or in need of correction. It does not claim that BOMA is superior to any existing foundational system. It does not claim to resolve the philosophical questions surrounding foundational pluralism. It does not claim that the applications surveyed in §7 have been realized. The claim is more modest: that there is an architectural dimension of foundational mathematics that has been systematically invisible within the existing paradigm, that this dimension can be made precise, and that a framework for doing so can be developed with sufficient rigor.

## 1.6 Paper Organization

Section 2 characterizes the existing foundational paradigm. Section 3 develops BOMA in full. Section 4 introduces the Foundational Space. Section 5 develops the theory of structural comparison. Section 6 situates existing foundations within BOMA. Section 7 surveys applications and research programs. Section 8 concludes.

Several cross-cutting remarks developed in the course of the paper address issues that span multiple sections. Remark 3.35 introduces a three-level logical architecture. Remark 6.1 introduces architectural reconstruction as the interpretive methodology governing the analyses of §6. Remark 4.3 establishes the Foundational Space as a possibility space. Remark 8.1 articulates the full scope of the framework’s vision. These remarks are integral to the paper’s argument.

## 2 The Current Paradigm

### 2.1 The Structure of Existing Foundations

*Observation 2.1* (The Linear Presentation). Every major foundational system is presented as a *linear development* [10, 38]: a sequence of declarations ordered so that each element presupposes only what has been declared before it. The sequence begins at a fixed starting point and extends forward without branching.

A compatible linearization of the Block dependency structure — as defined in Definition 3.15 — is precisely what a traditional textbook presentation provides: one admissible reading order among many. Different textbooks on the same foundational system may select different linearizations, but all are compatible with the same underlying dependency structure.

*Observation 2.2* (Primitive Notions and Axioms as Fixed Points). In every existing foundational system, the primitive notions and initial axioms are presented as fixed points: given without derivation and treated as the terminal points of any regress of justification.

### 2.2 What Existing Foundations Achieve

*Observation 2.3* (Consistency and Coherence). The primary achievement of existing foundational systems is the provision of a framework within which mathematical reasoning can be conducted without risk of contradiction [12, 13].

*Observation 2.4* (Expressive Power). Existing foundational systems are expressively powerful to an extent that continues to exceed the demands placed on them by mathematical practice [20, 27].

*Observation 2.5* (Computability and Mechanization). The formal specification of existing foundations has enabled the development of proof assistants. Coq [40], Lean [31], Agda [33], Isabelle [32], and Metamath [29] are among the systems that have made this possible.

*Observation 2.6* (Theoretical Comparability via Interpretability). Existing foundational systems have been compared through the notion of relative interpretability [38, 22].

## 2.3 The Architectural Gap

*Observation 2.7* (The Invisibility of Decision Points). Every existing foundational system passed through moments of genuine choice. These moments are not invented by the present framework. They are episodes in the history of mathematics, documented in primary sources [30, 17, 43]. What they share is a common structure: a moment at which more than one mathematically coherent path was available, a process of deliberation, and a selection. We term this structure a *Decision Point*.

The observation that motivates the present paper is not that existing formal systems fail to contain a concept we have introduced. It is that the formal presentations of existing systems provide no structural mechanism for recording, within the formalism itself, the Decision Points that their historical development passed through.

*Observation 2.8* (The Suppression of Alternatives). A consequence of Observation 2.7 is that the alternatives to the choices actually made are also invisible.

*Observation 2.9* (The Absence of Architectural History). The combination of invisible Decision Points and suppressed alternatives produces the *absence of architectural history*: the phenomenon in which a foundational system presents its current state without any record of how it came to be in that state.

*Observation 2.10* (The Conflation of Injections and Derivations). Existing foundational presentations do not distinguish between the assumptions they inject and the theorems they derive in terms of their architectural status.

## 2.4 The Nature of the Gap: Architectural, Not Logical

*Remark 2.11.* The gap identified in §2.3 is not a logical gap. Existing foundational systems are logically complete in the sense relevant to their purposes. The gap is *architectural*: existing foundations are organized according to a paradigm that optimizes for logical consistency, expressive power, and mechanizability — but not for transparency of Decision Points, preservation of alternatives, or explicit injection structure.

*Remark 2.12.* The architectural gap is not a defect in existing foundations; it is a dimension in which they are silent. The architecture proposed in this paper does not correct existing foundations; it extends the paradigm within which foundations are organized.



## 2.5 Existing Foundations Within the Architectural Paradigm

*Observation 2.13.* Every existing foundational system corresponds, within the architectural framework, to a locally consistent path in some Foundational Space  $\mathfrak{F}(\beta_0)$ .

*Observation 2.14.* Existing foundational systems correspond to *architecturally opaque* paths: paths that satisfy local consistency but do not satisfy Architectural Transparency, because their Decision Points are not documented and their Branch Trees are not recorded.

The application of the architectural framework to existing foundations is an instance of what §6 terms *architectural reconstruction*: a retrospective interpretive reading, not a mechanical extraction.

## 3 The Proposed Architecture (BOMA)

### 3.1 Blocks

#### Motivation

Every formal development of mathematics organizes its content into primitive notions, axioms, definitions, theorems, and inference rules. What is systematically absent is an explicit account of the *unit of packaging* — the structure that groups related content together, records what that content depends upon, and tracks whether it is stable or subject to revision. That structure is the *Block*.

**Definition 3.1** (Block). A *Block* is a four-tuple:

$$\beta = \left( \text{Id}(\beta), \text{Content}(\beta), \text{Dep}(\beta), \text{Status}(\beta) \right)$$

where:

- $\text{Id}(\beta)$  is a unique identifier from a totally ordered index set  $\mathcal{I}$ ;
- $\text{Content}(\beta)$  is a finite, non-empty collection of formal statements in a declared language  $\mathcal{L}$ ;
- $\text{Dep}(\beta) \subseteq \{\beta' \mid \text{Id}(\beta') < \text{Id}(\beta)\}$  is the set of Blocks directly presupposed by  $\text{Content}(\beta)$ ;
- $\text{Status}(\beta) \in \{\text{Fixed}, \text{Open}\}$ .

**Definition 3.2** (Fixed Block). A Block  $\beta$  is *Fixed* if its content is stable within the current architectural path. A Fixed Block is a reliable node; any Block depending on it may do so without anticipating changes.

**Definition 3.3** (Open Block). A Block  $\beta$  is *Open* if its content is subject to revision. An Open Block may transition to Fixed once its content is stabilized. The reverse transition is not permitted within a given path; revision is effected by branching, not by modifying in place.

**Definition 3.4** (Atomic Block). A Block  $\beta$  is *Atomic* if  $\text{Dep}(\beta) = \emptyset$ . Atomic Blocks are the roots of the architectural development, carrying primitive notions, initial axioms, and the declaration of the underlying logic.

*Remark 3.5.* Every architectural development must contain at least one Atomic Block. The Atomic Block is the location at which the most fundamental choices are made explicit: what counts as a primitive notion, what logic governs inference, what is accepted without derivation. In traditional foundational presentations, these choices are made implicitly. In BOMA, they are the declared content of the first Block.

**Definition 3.6** (Block Identity and Content-Equivalence). Two Blocks  $\beta$  and  $\beta'$  are *identical* if and only if  $\text{Id}(\beta) = \text{Id}(\beta')$ . They are *content-equivalent* if  $\text{Content}(\beta) = \text{Content}(\beta')$  up to notational equivalence.

*Remark 3.7* (Blocks and Theory Structuring). The notion of a Block as defined in Definition 3.1 is related to, but distinct from, the notion of a structured theory presentation in algebraic specification languages such as CASL [36] and in mathematical knowledge management systems such as OMDOC [21] and MMT [35]. In these systems, a theory extension records what was added; a Block in BOMA records, additionally, the injection structure of the added content (§3.6) and — when the Block is a Decision Block — the alternatives that were not chosen and the rationale for the selection (§3.3).

## 3.2 Dependencies and Ordering

**Definition 3.8** (Direct Dependency).  $\beta'$  *directly depends on*  $\beta$ , written  $\beta \prec_1 \beta'$ , if and only if  $\beta \in \text{Dep}(\beta')$ .

**Definition 3.9** (Dependency Relation). The *dependency relation*  $\prec$  is the transitive closure of  $\prec_1$ . We say  $\beta$  is an *ancestor* of  $\beta'$  and  $\beta'$  is a *descendant* of  $\beta$ .

**Definition 3.10** (Dependency Closure). The *dependency closure* of  $\beta$  is:

$$\overline{\text{Dep}}(\beta) = \{ \beta' \mid \beta \prec \beta' \}.$$

**Axiom 3.11** (Acyclicity). The dependency relation  $\prec$  is irreflexive and acyclic.

**Proposition 3.12.** *If  $\prec$  is acyclic on a finite set of Blocks, then it is well-founded: every non-empty subset contains at least one Block with no ancestors within that subset. The minimal elements are precisely the Atomic Blocks of Definition 3.4.*

*Proof.* Let  $\{\beta_0, \dots, \beta_n\}$  be a finite set of Blocks with an acyclic dependency relation  $\prec$ . Consider any non-empty subset  $S$ . Since  $S$  is finite and  $\prec$  is acyclic on  $S$ , there are no infinite descending chains. A finite acyclic relation is well-founded, so every non-empty finite set with an irreflexive transitive relation contains at least one element with no predecessors within that set. Such  $\beta^* \in S$  satisfies: there is no  $\beta' \in S$  with  $\beta' \prec \beta^*$ . Taking  $S$  to be the full set of Blocks, the minimal elements are those with no predecessors in the entire path, i.e., Blocks  $\beta$  for which  $\text{Dep}(\beta) = \emptyset$ . These are precisely the Atomic Blocks.  $\square$

**Proposition 3.13.** *The dependency relation  $\prec$  together with the identity relation defines a strict partial order on the Blocks of any architectural path.*

*Proof.* Irreflexivity follows from Axiom 3.11. Transitivity holds by construction. For asymmetry: suppose  $\beta \prec \beta'$  and  $\beta' \prec \beta$ . By transitivity of  $\prec$ , we obtain  $\beta \prec \beta$ , contradicting irreflexivity. Hence  $\prec$  is asymmetric, and together with irreflexivity and transitivity, constitutes a strict partial order.  $\square$

*Remark 3.14.* The structure is a *partial* order, not a total one. In a partial order, several Blocks may be mutually independent and developed in parallel. This reflects the actual structure of mathematical knowledge more faithfully than a linear presentation.

**Definition 3.15** (Compatible Linearization). A *compatible linearization* of the Blocks of a path is a total order  $<_\ell$  such that  $\beta \prec \beta' \implies \beta <_\ell \beta'$ . Compatible linearizations exist for every finite acyclic dependency structure (standard topological sorting theorem [10]).

**Definition 3.16** (Cumulative Content). The *cumulative content* of  $\beta$  is:

$$\widehat{\text{Content}}(\beta) = \text{Content}(\beta) \cup \bigcup_{\beta' \in \overline{\text{Dep}}(\beta)} \text{Content}(\beta').$$

### 3.3 Decision Points

#### Motivation

Every foundational development contains moments of genuine choice. The architecture requires that every genuine choice be made explicit, recorded at the moment it is made, and preserved as a permanent feature of the architectural path.

**Definition 3.17** (Decision Point). A *Decision Point* is a five-tuple:

$$\mathcal{D} = (\text{Id}(\mathcal{D}), \text{Anchor}(\mathcal{D}), \text{Options}(\mathcal{D}), \text{Selection}(\mathcal{D}), \text{Rationale}(\mathcal{D}))$$

where  $|\text{Options}(\mathcal{D})| \geq 2$ .

*Remark 3.18.* The Rationale component is not required to be a formal proof that the selection is superior to the alternatives. It is a transparent account of the considerations that guided the selection — mathematical, pragmatic, historical, or a combination.

**Axiom 3.19** (Mandatory Documentation). Every Decision Point must be fully documented at the moment it is introduced:

1.  $\text{Options}(\mathcal{D})$  enumerates all genuinely considered alternatives;
2.  $\text{Selection}(\mathcal{D})$  is unambiguous;
3.  $\text{Rationale}(\mathcal{D})$  is explicit;
4. Unchosen options are permanently preserved.

**Definition 3.20** (Forced Step and Genuine Choice). A step from Block  $\beta$  is *forced* if the content of the next Block is uniquely determined by prior cumulative content. A step is a *genuine choice* — objectively, if the alternatives’ logical admissibility and non-equivalence is established by a formal independence result; *interpretively*, if the developer judges it so without a formal independence proof.

*Remark 3.21* (The Interpretive Character of Decision Point Identification). The distinction between forced steps and genuine choices is not always decidable from within the formal record of the path. Both objectively and interpretively identified Decision Points are legitimate, but they carry different epistemic weights. The basis of identification must be declared in the Rationale component of the Decision Block.

This interpretive dimension is not a deficiency of the framework. It reflects an irreducible feature of foundational construction: the assessment of what is genuinely optional is itself a foundational act. The framework does not eliminate interpretive judgment; it makes it visible and auditable.

**Definition 3.22** (Decision Block). A *Decision Block* is a Block whose content is precisely the documentation of a Decision Point satisfying Axiom 3.19.

**Definition 3.23** (Decision Point Catalogue). The *Decision Point Catalogue* of an architectural path is the ordered collection of all Decision Blocks, ordered by position in the dependency structure. It constitutes the complete record of every genuine choice made in the development.

*Remark 3.24.* In existing foundational systems, the analogue of a Decision Point Catalogue does not exist as an explicit artifact. The choices that shaped ZFC, Peano Arithmetic, or the Calculus of Constructions are recoverable only through historical scholarship. BOMA makes the catalogue a first-class structural object.

### 3.4 Branches

**Definition 3.25** (Path). An *architectural path*  $\mathcal{P}$  is a finite sequence of Blocks  $(\beta_0, \beta_1, \dots, \beta_n)$  satisfying:

1.  $\beta_0$  is Atomic;
2. For each  $i > 0$ ,  $\text{Dep}(\beta_i) \subseteq \{\beta_0, \dots, \beta_{i-1}\}$ ;
3. The dependency relation is acyclic (Axiom 3.11);
4. Every Decision Block satisfies Axiom 3.19.

The local consistency requirement referenced here is defined formally in §3.5. This forward dependency is unavoidable given that the validity criterion of §3.5 presupposes the path structure of this section; the two sections are mutually dependent in their motivation and should be read as a unified specification.

**Definition 3.26** (Trunk Path). The *Trunk Path*  $\mathcal{P}_0$  is the initial architectural path — the path produced by following, at every Decision Point, the option recorded as  $\text{Selection}(\mathcal{D})$ .

**Definition 3.27** (Branching Prefix). Let  $\mathcal{P}$  contain a Decision Block  $\beta_j$  for  $\mathcal{D}$ . The *branching prefix* of  $\mathcal{P}$  at  $\mathcal{D}$  is the initial segment  $\mathcal{P}_{|\mathcal{D}} = (\beta_0, \dots, \beta_j)$ .

**Definition 3.28** (Branch). Let  $\mathcal{P}$  contain Decision Point  $\mathcal{D}$  with options  $\{O_1, \dots, O_k\}$  and selection  $O_s$ . A *Branch* at  $\mathcal{D}$  selecting option  $O_i \neq O_s$  is an architectural path:

$$\mathcal{B}_i = (\beta_0, \dots, \beta_j, \beta_{j+1}^i, \beta_{j+2}^i, \dots)$$

where  $(\beta_0, \dots, \beta_j) = \mathcal{P}_{|\mathcal{D}}$  is the branching prefix inherited exactly, and subsequent Blocks are developed under the commitment of  $O_i$ .

**Definition 3.29** (Branch at Depth  $d$ ). A Branch at the  $m$ -th Decision Point (ordered by position in the dependency structure) is said to be at *depth*  $m$ . Branches at greater depth share a longer common ancestry and inherit a more developed foundational context before diverging.

**Proposition 3.30** (Inheritance of Branching Prefix). *Let  $\mathcal{B}_i$  be a Branch at  $\mathcal{D}$  with branching prefix  $(\beta_0, \dots, \beta_j)$ . Then every Block  $\beta_\ell$  for  $\ell \leq j$  appears in  $\mathcal{B}_i$  with the same identifier, content, dependency set, and status as in  $\mathcal{P}$ ; and  $\widehat{\text{Content}}(\beta_j)$  is identical in both.*

*Proof.* Immediate from Definition 3.28: the branching prefix is inherited exactly, with no modification to any Block’s components.  $\square$

**Principle 3.31** (Branch Independence). The development of  $\mathcal{B}_i$  after  $\mathcal{D}$  is fully independent of  $\mathcal{P}$  after  $\mathcal{D}$ : no post-Decision Block of  $\mathcal{P}$  is available to  $\mathcal{B}_i$  unless independently derived within  $\mathcal{B}_i$ ; and no result of  $\mathcal{B}_i$  affects or constrains  $\mathcal{P}$ .

**Definition 3.32** (Branch Tree). The *Branch Tree*  $\mathcal{T}$  of a development is the rooted tree whose root is the Atomic Block  $\beta_0$ , whose internal nodes are Decision Blocks, whose edges are labeled by the options  $O_i \in \text{Options}(\mathcal{D})$ , and whose paths from root to any node correspond to unique architectural paths.

*Remark 3.33.* The Foundational Space  $\mathfrak{F}(\beta_0)$  defined formally in §4 is the collection of all locally consistent paths sharing  $\beta_0$ . The Branch Tree is the architectural record of one development within that space.

**Definition 3.34** (Minimal Branch). A Branch  $\mathcal{B}_i$  at  $\mathcal{D}$  is *minimal* if the first Post-Decision Block differs from the corresponding Block in the parent path only in the content directly determined by the selection of  $O_i$ , and all subsequent Blocks are developed under no additional assumptions beyond the branching prefix and the commitment of  $O_i$ .

### 3.5 Local Consistency as the Sole Validity Criterion

*Remark 3.35* (Three Logical Levels). The architecture operates simultaneously at three logical levels.

The *object logic* is declared within each individual path at its Atomic Block. The architecture is genuinely indifferent to the choice of object logic.

The *framework logic* is the logic used to define the architectural concepts themselves. The present paper employs classical first-order logic [10, 38] as its framework logic. This is a choice, not a necessity: a version of the framework formulated in intuitionistic logic would be a coherent alternative, constituting a Branch from the present formulation at the Decision Point of framework logic selection.

The *meta-logic* is the logic in which the propositions and proofs of this paper are expressed; it is likewise classical.

The claim of logical neutrality applies exclusively to the object logic level.

**Definition 3.36** (Deductive Closure of a Branch). The *deductive closure* of  $\mathcal{B}$  is:

$$\text{Th}(\mathcal{B}) = \{ \varphi \mid \mathcal{B} \vdash \varphi \}$$

where  $\vdash$  denotes derivability under the object logic declared in the Atomic Block of  $\mathcal{B}$ .

**Definition 3.37** (Local Consistency). A branch  $\mathcal{B}$  is *locally consistent* if there is no formula  $\varphi$  in the language of the path's declared object logic such that both  $\mathcal{B} \vdash \varphi$  and  $\mathcal{B} \vdash \neg\varphi$ , where  $\neg$  denotes negation as defined within that object logic.

*Remark 3.38.* The use of  $\neg\varphi$  in Definition 3.37 is relative to the object logic of the path. For logics without a standard negation, an appropriate analogue must be declared in the Atomic Block.

**Definition 3.39** (Branch Validity). A branch  $\mathcal{B}$  is *valid* if and only if it is locally consistent. No further condition is imposed.

*Remark 3.40* (On the Use of “Principle”). Throughout this paper, the term *Principle* designates an architectural commitment that is neither a primitive assumption (Axiom) nor a derived result (Proposition), but a structural norm: a requirement that follows from the design goals of the framework and governs how its components are to be used.

*Remark 3.41* (Minimality of the Validity Criterion). Branch validity requires neither consistency with any external system, nor agreement with any other branch, nor completeness. Local consistency is both necessary and sufficient.

**Principle 3.42** (Independence of Validity). The local consistency of  $\mathcal{B}_1$  is logically independent of the local consistency of  $\mathcal{B}_2$ . If  $\mathcal{B}_1 \vdash \varphi$  and  $\mathcal{B}_2 \vdash \neg\varphi$ , both branches may nonetheless be locally consistent, and therefore both valid.

**Principle 3.43** (Contradiction as Architectural Information). Let  $\mathcal{B}_1$  and  $\mathcal{B}_2$  be valid branches with:

$$\mathcal{B}_1 \vdash \varphi \quad \text{and} \quad \mathcal{B}_2 \vdash \neg\varphi.$$

This situation is not a failure of the architecture. It is the precise content the architecture is designed to surface: the Decision Point  $\mathcal{D}$  at which they diverged was *non-trivial*. Both mathematical worlds are coherent on their own terms, and the architecture preserves both without forcing a resolution.

**Definition 3.44** (Non-trivial Decision Point). A Decision Point  $\mathcal{D}$  is *non-trivial* if there exist two valid branches  $\mathcal{B}_1$  and  $\mathcal{B}_2$  diverging at  $\mathcal{D}$  such that for some formula  $\varphi$ :  $\mathcal{B}_1 \vdash \varphi$  and  $\mathcal{B}_2 \vdash \neg\varphi$ .

*Remark 3.45.* Comparison between branches is addressed to two or more branches: what do their differences reveal? This is an epistemological question, not a logical one. Comparison is a tool for learning, not for adjudication. There is no mechanism by which one branch can invalidate another.

### 3.6 Minimal Injection

**Definition 3.46** (Injection). A statement  $\varphi \in \text{Content}(\beta)$  is an *injection* of  $\beta$  if:

$$\widehat{\text{Content}}(\overline{\text{Dep}}(\beta)) \not\vdash \varphi.$$

We write:

$$\text{Inj}(\beta) = \left\{ \varphi \in \text{Content}(\beta) \mid \widehat{\text{Content}}(\overline{\text{Dep}}(\beta)) \not\vdash \varphi \right\}, \quad \text{Der}(\beta) = \text{Content}(\beta) \setminus \text{Inj}(\beta).$$

**Definition 3.47** (Injection Weight). The *injection weight* of  $\beta$  is  $\omega(\beta) = |\text{Inj}(\beta)|$ .

**Axiom 3.48** (Minimal Injection Principle). Every Block  $\beta$  must satisfy:

1. **Necessity:** every injection is necessary for the stated purpose of  $\beta$ ;
2. **Individuation:** every injection is declared individually and explicitly;
3. **Non-derivability Acknowledgment:** the fact that  $\varphi$  is an injection is acknowledged;
4. **Minimality:** if  $\varphi$  can be replaced by a logically weaker  $\psi$  sufficient for the purpose of  $\beta$ , then  $\psi$  is injected instead.

*Remark 3.49* (Injection Weight and Computability). The injection set  $\text{Inj}(\beta)$  is well-defined but subject to an important qualification. In the general case of first-order logic, determining whether  $\varphi$  is derivable from a set of prior statements is undecidable, as an immediate consequence of Church's theorem [7]. The injection set is therefore well-defined but not mechanically computable in full generality.

This qualification does not undermine the architectural significance of injection weights. In many practically relevant cases — finitely axiomatized systems with decidable consequence relations — injection weights are effectively computable. The value of the Minimal Injection Principle lies in the architectural discipline it enforces: the developer is required to identify and declare each injection individually. This discipline is independent of whether a mechanical verifier could confirm the non-derivability claim.



**Definition 3.50** (Cumulative Injection Weight). The *cumulative injection weight* of a path  $\mathcal{P} = (\beta_0, \dots, \beta_n)$  is:

$$\Omega(\mathcal{P}) = \sum_{i=0}^n \omega(\beta_i).$$

**Proposition 3.51.** *Let  $\beta$  satisfy the Minimal Injection Principle, and let  $\varphi \in \text{Inj}(\beta)$  be such that there exists an alternative statement  $\psi$  with  $\widehat{\text{Content}(\overline{\text{Dep}(\beta)})} \not\vdash \psi$ ,  $\varphi \not\vdash \psi$ , and  $\psi \not\vdash \varphi$ . Then  $\beta$  is associated with a genuine choice requiring a Decision Block.*

*Proof.* Since  $\psi$  is not derivable from prior content, injecting  $\psi$  in place of  $\varphi$  represents a distinct foundational commitment. The path  $\mathcal{P}_\psi$  injecting  $\psi$  and the path  $\mathcal{P}_\varphi$  injecting  $\varphi$  are both locally consistent by the Minimal Injection Principle. Since  $\varphi \not\vdash \psi$  and  $\psi \not\vdash \varphi$ , the two paths represent genuinely different foundational commitments. By Definition 3.20, this is a genuine choice. By Axiom 3.19, it must be documented.

In the case where  $\mathcal{P}_\psi \vdash \neg\varphi$ , the Decision Point is non-trivial in the sense of Definition 3.44. In the general case where  $\mathcal{P}_\psi \not\vdash \neg\varphi$ , the Decision Point is still a genuine choice — two locally consistent paths diverge with different foundational commitments — and must be documented, even if it does not meet the stronger condition of Definition 3.44.  $\square$

*Remark 3.52* (Minimal Injection Applied to Logical Commitments). The Minimal Injection Principle applies not only to mathematical content but to logical commitments. A logically minimal development defaults to intuitionistic logic and introduces classical commitments only when necessary, since intuitionistic logic is strictly weaker than classical logic. Per-injection logic declarations — specifying the inference system under which each injection is introduced — are the logical analogue of the per-injection necessity declarations required by Axiom 3.48.

### 3.7 Architectural Transparency

**Definition 3.53** (Dependency Legibility). A path  $\mathcal{P}$  is *dependency-legible* if, for every Block  $\beta \in \mathcal{P}$ , the set  $\text{Dep}(\beta)$  is explicitly declared, and  $\overline{\text{Dep}(\beta)}$  is reconstructible from these declarations without inference or supplementation.

**Definition 3.54** (Decision Legibility). A path  $\mathcal{P}$  is *decision-legible* if every Decision Point is documented in a Decision Block satisfying Axiom 3.19.

*Remark 3.55* (Transparency of Decision Point Basis). Decision Legibility requires not only that Decision Points be documented but that the basis of their identification — objective or interpretive, in the sense of Definition 3.20 — be declared in the Rationale component. A development in which Decision Points are recorded but their basis of identification is suppressed presents interpretive judgments as if they were formally determined.

**Definition 3.56** (Injection Legibility). A path  $\mathcal{P}$  is *injection-legible* if, for every Block  $\beta$ , the injection set  $\text{Inj}(\beta)$  is explicitly identified, every element is individually declared as an injection, and the path satisfies Axiom 3.48.

**Definition 3.57** (Branch Legibility). A path  $\mathcal{P}$  is *branch-legible* if the Branch Tree  $\mathcal{T}$  is fully documented: every Branch is identified by its originating Decision Point, its founding selection, and its branching prefix; and the local consistency status of every Branch is declared.

**Definition 3.58** (Architectural Transparency). A path  $\mathcal{P}$  is *architecturally transparent* if and only if it is dependency-legible, decision-legible, injection-legible, and branch-legible.

*Remark 3.59.* Architectural Transparency is a conjunctive property: satisfying three of the four conditions without the fourth yields a development that is only partially auditable.

**Proposition 3.60.** *If  $\mathcal{P}$  is architecturally transparent, then every initial segment  $(\beta_0, \dots, \beta_k)$  for  $k \leq n$  is also architecturally transparent.*

*Proof.* Each of the four legibility conditions applies to individual Blocks and their declared structural components. Dependency and Injection Legibility hold for initial segments trivially. Decision Legibility holds since every Decision Block in the initial segment satisfies Axiom 3.19. Branch Legibility holds since the Branch Tree of the initial segment is a sub-tree of the full Branch Tree, and the documentation requirement for the initial segment is satisfied as a restriction of the full documentation.  $\square$

*Remark 3.61* (Architectural Transparency and Institution Theory). Institution theory [15] achieves a form of logical neutrality at the mathematical level by abstracting over the relationship between sentences and models across different logical systems. BOMA achieves neutrality by requiring that the object logic be declared as a component of the Atomic Block. These are different architectural strategies for the same goal. Institution theory characterizes what is logically common across systems; BOMA characterizes what is architecturally required of any system, regardless of its logical character. The two frameworks are complementary.

*Remark 3.62* (Consolidated Statement of the Architecture). An architecturally transparent foundational development is a Branch Tree  $\mathcal{T}$  whose paths are sequences of Blocks satisfying: explicit Block declarations (§3.1); acyclic dependency partial order (§3.2); fully documented Decision Points (§3.3); independently developing Branches (§3.4); local consistency as the sole validity criterion (§3.5); Minimal Injection for every Block (§3.6); and full documentation available to any reader without reconstruction (§3.7).

## 4 Branching and the Space of Foundations

### 4.1 The Space of Foundations

**Definition 4.1** (Foundational Space). Let  $\beta_0$  be a fixed Atomic Block. The *Foundational Space* relative to  $\beta_0$ , written  $\mathfrak{F}(\beta_0)$ , is the collection of all locally consistent architectural paths whose first element is  $\beta_0$ :

$$\mathfrak{F}(\beta_0) = \{ \mathcal{P} \mid \mathcal{P} \text{ is a locally consistent architectural path with } \mathcal{P}_0 = \beta_0 \}.$$

*Remark 4.2.* The Foundational Space is not merely a catalogue of existing foundational developments [5, 17]. It is a *possibility space*: the collection of all locally consistent architectural paths, the vast majority of which have not been traversed by any mathematical development.

*Remark 4.3* (The Foundational Space as Possibility Space). Unknown mathematics is not, in this picture, mathematics that exists somewhere waiting to be found, nor mathematics that does not exist until invented. It is mathematics that *could be* — paths in  $\mathfrak{F}(\beta_0)$  whose Blocks are determinable, whose Decision Points are documentable, and whose local consistency is verifiable, but whose content has not yet been produced by any development.

Every non-trivial Decision Point in an existing development is a bifurcation: the path taken and at least one path not taken. Every unchosen option at every documented Decision Point is an entrance to unknown mathematics. BOMA does not merely organize what is known. It makes the structure of what is not yet known visible, navigable, and methodologically tractable.

*Remark 4.4* (Relation to the Set-Theoretic Multiverse). The Foundational Space  $\mathfrak{F}(\beta_0)$  bears a structural resemblance to the set-theoretic multiverse proposed by Hamkins [16], and the relationship between the two frameworks merits explicit discussion.

Hamkins argues that the independence results of set theory motivate a pluralist ontological stance: there is no single “true” set-theoretic universe, but rather a multiverse of equally legitimate universes, each with its own truths.

The Foundational Space  $\mathfrak{F}(\beta_0)$  differs from Hamkins’ multiverse in five fundamental respects.

*Scope.* Hamkins’ multiverse is exclusively set-theoretic. The Foundational Space accommodates paths of any foundational character — set-theoretic, type-theoretic, categorical, or otherwise. The set-theoretic paths in  $\mathfrak{F}(\beta_0)$  constitute a specific cluster analogous to Hamkins’ multiverse, but a proper substructure of the full Foundational Space.

*Level of operation.* Hamkins' multiverse operates at the level of mathematical models, with relationships established by forcing and inner models. The Foundational Space operates at the level of architectural construction, with relationships established by common ancestry, divergence points, and Decision Point reselections.

*Ontological commitment.* Hamkins' multiverse carries an explicit ontological commitment: the plurality of universes is a claim about what genuinely exists. The Foundational Space carries no such commitment.

*Documentation.* The universes of Hamkins' multiverse exist independently of any documentation of the decisions that produced them. In contrast, a path in  $\mathfrak{F}(\beta_0)$  is an architecturally transparent development with explicitly documented Decision Points and injection structure.

*Mechanism of plurality.* In Hamkins' multiverse, plurality arises from model-theoretic operations within set theory itself. In the Foundational Space, plurality arises from architectural decisions that precede mathematical development.

In summary: Hamkins' multiverse can be understood as a specific, densely explored region of the Foundational Space — the cluster of set-theoretic paths diverging at Decision Points internal to ZFC and its extensions — studied through the lens of model-theoretic semantics rather than architectural construction.

## 4.2 Structure of the Foundational Space

**Definition 4.5** (Common Ancestry). The *common ancestry* of  $\mathcal{P}$  and  $\mathcal{Q}$ , written  $\mathcal{P} \wedge \mathcal{Q}$ , is the longest initial segment shared by both paths.

**Definition 4.6** (Divergence Point). The *divergence point* of  $\mathcal{P}$  and  $\mathcal{Q}$  is the Decision Point  $\mathcal{D}$  immediately following their common ancestry.

**Proposition 4.7** (Architectural Ultra-Pseudometric). Define  $d : \mathfrak{F}(\beta_0) \times \mathfrak{F}(\beta_0) \rightarrow \mathbb{R}_{\geq 0}$  by:

$$d(\mathcal{P}, \mathcal{Q}) = \frac{1}{|\mathcal{P} \wedge \mathcal{Q}| + 1},$$

where  $|\mathcal{P} \wedge \mathcal{Q}|$  is taken as  $\infty$  when the common ancestry is infinite, with convention  $\frac{1}{\infty+1} = 0$ . Then  $d$  satisfies non-negativity, symmetry, and the ultrametric inequality, constituting an ultra-pseudometric on  $\mathfrak{F}(\beta_0)$ .

*Proof.* Non-negativity and symmetry are immediate. For the ultrametric inequality: let  $j = |\mathcal{P} \wedge \mathcal{Q}|$  and  $k = |\mathcal{Q} \wedge \mathcal{R}|$ , and set  $m = \min(j, k)$ . Since  $\mathcal{P}$  and  $\mathcal{Q}$  agree on their first  $j$  Blocks, and  $\mathcal{Q}$  and  $\mathcal{R}$  agree on their first  $k$  Blocks, both  $\mathcal{P}$  and  $\mathcal{R}$  agree with  $\mathcal{Q}$  on their first  $m$  Blocks, hence agree with each other on those  $m$  Blocks. Therefore  $|\mathcal{P} \wedge \mathcal{R}| \geq m = \min(j, k)$ , giving:

$$d(\mathcal{P}, \mathcal{R}) \leq \frac{1}{\min(j, k) + 1} = \max\left(\frac{1}{j + 1}, \frac{1}{k + 1}\right) = \max(d(\mathcal{P}, \mathcal{Q}), d(\mathcal{Q}, \mathcal{R})). \quad \square$$

*Remark 4.8* (Packaging Dependence and Its Interpretation). The ultra-pseudometric  $d$  depends on  $|\mathcal{P} \wedge \mathcal{Q}|$  — the Block count of the common ancestry — rather than on any intrinsic measure of logical content. The same logical content packaged into more Blocks yields a smaller distance.

This dependence is a feature, not a defect, provided its interpretation is correctly understood. The function  $d$  measures *architectural proximity* — the degree to which two paths proceeded in documented agreement — rather than *logical proximity*. Architectural proximity is the appropriate measure in the context of BOMA, which is concerned with the process of foundational construction and not only with the resulting content.

*Remark 4.9* (Pseudometric Character). The function  $d$  fails to be a genuine metric when infinite paths are admitted, since two paths sharing an infinite common ancestry receive distance zero without being identical. This correctly reflects architectural indistinguishability at any finite depth. For all practical purposes — finite foundational developments — the restriction to finite-length paths gives a genuine ultrametric.

*Remark 4.10* (Canonical Normalization). When the primary purpose of computing  $d$  is to compare the logical depth of shared commitment, paths may be normalized to a common level of granularity before computing the Block count. A *canonical normalization* re-packages content into Blocks of uniform logical weight (one injection per Block). The normalized distance  $d^*$  is invariant under re-packaging and measures shared logical commitment. The relationship between  $d$  and  $d^*$  is itself architecturally informative.

Canonical normalization in §5.5 provides a basis for comparison invariant under packaging decisions.

**Definition 4.11** (Ancestry Depth). The *ancestry depth* of  $\mathcal{P}$  and  $\mathcal{Q}$  is  $\delta(\mathcal{P}, \mathcal{Q}) = |\mathcal{P} \wedge \mathcal{Q}|$ .

### 4.3 Navigation in the Foundational Space

**Definition 4.12** (Reachability). A path  $\mathcal{Q}$  is *reachable* from  $\mathcal{P}$  if there exists a finite sequence of Decision Point reselections connecting them. Since the Branch Tree is connected, every pair of paths is mutually reachable.

### 4.4 Contradictions as Structural Information

**Definition 4.13** (Inter-path Contradiction). An *inter-path contradiction* relative to  $\mathcal{D}$  is a formula  $\varphi$  such that  $\mathcal{P} \vdash \varphi$  and  $\mathcal{Q} \vdash \neg\varphi$  for branches  $\mathcal{P}, \mathcal{Q}$  diverging at  $\mathcal{D}$ .

**Theorem 4.14** (Contradiction Localization). *Let  $\mathcal{P}$  and  $\mathcal{Q}$  be locally consistent paths with  $\mathcal{P} \vdash \varphi$  and  $\mathcal{Q} \vdash \neg\varphi$ . Then:*

1. Neither  $\mathcal{P}$  nor  $\mathcal{Q}$  is internally inconsistent.
2. The inter-path contradiction is fully localized to the divergence point  $\mathcal{D}$ .
3. The inter-path contradiction certifies that  $\mathcal{D}$  is non-trivial.

*Proof.* (1) Both paths are locally consistent by hypothesis.

(2) By Proposition 3.30, the cumulative content of  $\mathcal{P} \wedge \mathcal{Q}$  is identical in both paths.

*That  $\varphi$  is not derivable from common ancestry:* Suppose  $\mathcal{P} \wedge \mathcal{Q} \vdash \varphi$ . Then both paths derive  $\varphi$ , but  $\mathcal{Q} \vdash \neg\varphi$ , so  $\mathcal{Q}$  derives both  $\varphi$  and  $\neg\varphi$ , contradicting local consistency of  $\mathcal{Q}$ .

*That  $\neg\varphi$  is not derivable from common ancestry:* Suppose  $\mathcal{P} \wedge \mathcal{Q} \vdash \neg\varphi$ . Then both paths derive  $\neg\varphi$ , but  $\mathcal{P} \vdash \varphi$ , so  $\mathcal{P}$  derives both  $\varphi$  and  $\neg\varphi$ , contradicting local consistency of  $\mathcal{P}$ .

Hence the derivations must depend on post-Decision content originating at  $\mathcal{D}$ .

(3) Immediate from Definition 3.44. □

**Definition 4.15** (Separation Formula and Semantic Signature). A *separation formula* for  $\mathcal{D}$  is a formula  $\varphi \in \text{Sep}(\mathcal{D})$  such that  $\mathcal{P} \vdash \varphi$  and  $\mathcal{Q} \vdash \neg\varphi$  for two locally consistent branches diverging at  $\mathcal{D}$ . The set  $\text{Sep}(\mathcal{D})$  is the *semantic signature* of  $\mathcal{D}$ .

## 4.5 Mathematics as a Traceable Knowledge System

**Definition 4.16** (Traceable Knowledge System). A foundational development is a *traceable knowledge system* if every statement in its deductive closure can be assigned a *provenance* — a complete record of the Blocks and Decision Point selections from which it ultimately derives — and this provenance is structurally determined from the architectural record.

**Proposition 4.17.** *Every architecturally transparent path in the sense of Definition 3.58 is a traceable knowledge system.*

*Proof.* Architectural Transparency requires Dependency Legibility, Injection Legibility, and Decision Legibility. These conditions ensure that the dependency closure of every Block, the injection set of every Block, and the selection at every Decision Point are explicitly declared. A derivation of  $\varphi$  in  $\mathcal{P}$  proceeds through a finite sequence of Blocks; its provenance is the subgraph of the dependency structure supporting this derivation, together with the injections and Decision Point selections on which it depends. This provenance is structurally determined — a definite object fixed by the architectural record. □

*Remark 4.18* (On Computability and Structural Determination). The claim is that provenance is *structurally determined*, not that it is *mechanically extractable* in all cases. Extraction depends on the decidability properties of the underlying logic. In cases where the logic has a decidable consequence relation, provenance extraction is mechanically feasible. In the general first-order case, it requires non-mechanical means.

## 5 Comparison as an Epistemological Tool

### 5.1 What Comparison Is Not

**Definition 5.1** (Adjudicative Comparison). An *adjudicative comparison* of two paths produces a judgment of the form:  $\mathcal{P}$  is preferable to  $\mathcal{Q}$ , where preferability is defined in terms of validity, correctness, or foundational superiority.

**Principle 5.2** (Prohibition on Adjudicative Comparison). BOMA provides no mechanism for adjudicative comparison. No structural feature — not the injection weight, not the depth of common ancestry, not the cardinality of the semantic signature — constitutes a criterion of foundational superiority between locally consistent paths.

### 5.2 What Comparison Is

**Definition 5.3** (Structural Comparison). A *structural comparison* of  $\mathcal{P}$  and  $\mathcal{Q}$  produces: the common ancestry; the divergence point; the selections made at the divergence point in each path; an approximation of the semantic signature; the injection weights of post-Decision Blocks; and the local consistency status of each path. No output includes a judgment of preferability.

**Definition 5.4** (Epistemological Yield). The *epistemological yield* of a structural comparison is the knowledge produced about the architecture of the Foundational Space — specifically about the nature of the divergence point, the mathematical consequences of the selection, and the relationship between the paths within  $\mathfrak{F}(\beta_0)$ .

### 5.3 Structural Ancestry and the Locus of Difference

**Proposition 5.5** (Difference Localization). *Every mathematical difference between  $\mathcal{P}$  and  $\mathcal{Q}$  is attributable to post-Decision content and localized to the divergence point  $\mathcal{D}$ .*

*Proof.* By Proposition 3.30, cumulative content of  $\mathcal{P} \wedge \mathcal{Q}$  is identical in both paths. Any statement derivable from common ancestry is derivable in both. Any disagree-

ment cannot derive from common ancestry and must depend on post-Decision content originating at  $\mathcal{D}$ .  $\square$

**Definition 5.6** (Injection Divergence). The *injection divergence* of  $\mathcal{P}$  and  $\mathcal{Q}$  with divergence point  $\mathcal{D}$  is:

$$\Delta_{\text{Inj}}(\mathcal{P}, \mathcal{Q}) = \left( \bigcup_{\beta \in \mathcal{P} \setminus \mathcal{P} \wedge \mathcal{Q}} \text{Inj}(\beta) \right) \Delta \left( \bigcup_{\beta \in \mathcal{Q} \setminus \mathcal{P} \wedge \mathcal{Q}} \text{Inj}(\beta) \right).$$

## 5.4 What We Learn from Contradictory Branches

**Theorem 5.7** (Epistemic Content of Contradiction). *Let  $\mathcal{P}$  and  $\mathcal{Q}$  be locally consistent paths with divergence point  $\mathcal{D}$  and separation formula  $\varphi \in \text{Sep}(\mathcal{D})$ . The structural comparison of  $\mathcal{P}$  and  $\mathcal{Q}$  with respect to  $\varphi$  yields:*

1. **Consistency confirmation:** *both paths are locally consistent;*
2. **Decision Point certification:**  *$\mathcal{D}$  is non-trivial;*
3. **Assumption identification:**  *$\Delta_{\text{Inj}}(\mathcal{P}, \mathcal{Q})$  contains the assumptions responsible for the different derivations;*
4. **World demarcation:**  *$\varphi$  is a statement whose truth value is determined by the selection at  $\mathcal{D}$ ;*
5. **Irreducibility confirmation:** *the disagreement is not resolvable within BOMA.*

*Proof.* Items (1) and (2) follow from Theorem 4.14. Item (3) follows from Definition 5.6 and Axiom 3.48: since every injection is individually declared and irredundant, assumptions responsible for any derived statement are identifiable from injection sets. Item (4) follows from Definition 4.15. Item (5) follows from Principle 5.2.  $\square$

## 5.5 Comparison Across Multiple Paths

**Definition 5.8** (Comparative Profile). Let  $\{\mathcal{P}_1, \dots, \mathcal{P}_m\}$  be locally consistent paths in  $\mathfrak{F}(\beta_0)$ . Their *comparative profile* consists of: for each pair, the common ancestry, divergence point, and ancestry depth; the pairwise distance matrix  $M_{ij} = d(\mathcal{P}_i, \mathcal{P}_j)$ ; for each Decision Point, the partition of paths by their selections; and for each separation formula, the partition of paths into those deriving  $\varphi$  and those deriving  $\neg\varphi$ .

Canonical normalization of paths — as defined in Remark 4.10 — provides a basis for comparison that is invariant under packaging decisions.



**Definition 5.9** (Foundational Cluster). A subset  $\mathcal{C} \subseteq \{\mathcal{P}_1, \dots, \mathcal{P}_m\}$  is a *foundational cluster at depth  $k$*  if  $\delta(\mathcal{P}_i, \mathcal{P}_j) \geq k$  for all  $\mathcal{P}_i, \mathcal{P}_j \in \mathcal{C}$ .

*Remark 5.10.* The ancestry depth of Definition 4.11 is related to the branch depth of Definition 3.29: two paths diverging at a Branch of depth  $d$  have ancestry depth equal to the number of Blocks in the branching prefix at that depth. Branches at greater depth share longer common ancestry.

## 5.6 Comparison as a Research Methodology

**Definition 5.11** (Architectural Inquiry). An *architectural inquiry* is a research program that applies structural comparison to a collection of paths with the goal of producing their comparative profile, identifying non-trivial Decision Points, characterizing their semantic signatures, and mapping the foundational clusters that emerge.

# 6 Relation to Existing Foundations

*Remark 6.1* (Architectural Reconstruction). The analyses of existing foundational systems in this section employ *architectural reconstruction*: the retrospective reading of an existing foundational development through the lens of BOMA, with the aim of identifying structures corresponding to Blocks, dependency relations, Decision Points, and injection sets.

Architectural reconstruction is an interpretive practice, not a mechanical extraction. The Decision Points identified below are not historical events in the sense of discrete, localized occurrences. They are logical structures — the existence of mutually consistent alternatives from which a selection was eventually made — not specific moments at which specific persons made specific choices.

The primary purpose of BOMA is prospective: it is designed for future foundational developments conducted within the architectural paradigm. The application to existing foundations is a secondary, illustrative purpose.

## 6.1 Zermelo-Fraenkel Set Theory with Choice

*Observation 6.2* (ZFC as an Architectural Path). ZFC [20, 44] corresponds to a locally consistent architectural path  $\mathcal{P}_{\text{ZFC}}$  in  $\mathfrak{F}(\beta_0^{\text{ZFC}})$ , where the Atomic Block declares: the membership relation  $\in$  as primitive; classical first-order logic; the universe of discourse consisting entirely of sets. Every axiom of ZFC corresponds to an injecting Block.

*Observation 6.3* (Decision-Point-Structured Episodes in ZFC). The historical development of ZFC contains several episodes with the logical structure of Decision Points.

*The Choice Episode.* Between 1904 and the mid-twentieth century [4, 30, 43], the mathematical community engaged in sustained deliberation about the Axiom of Choice. The episode involved explicit recognition of alternatives — Choice, Dependent Choice, Countable Choice, no choice principle — and explicit argument about their consequences. The architectural framework reads this as a Decision Point with semantic signature including the Well-Ordering Theorem [43], Zorn’s Lemma [20], and the Hahn-Banach Theorem [22].

*The Foundation Episode.* The Axiom of Foundation was introduced primarily as a technical device excluding pathological constructions. The alternative — non-well-founded sets — has been developed by Aczel and others [1, 2]. The reconstruction of this as a Decision Point is more tentative: the “choice” was partly a decision by default.

*The Classical Logic Episode.* The commitment to classical first-order logic was made against a background in which intuitionistic alternatives were known — Brouwer’s intuitionism [6] and Heyting’s formalization [18, 19] were contemporaneous.

*Remark 6.4.* The identification of Decision-Point-structured episodes in ZFC is an architectural reconstruction in the sense of Remark 6.1. It is an illustration of the framework’s concepts, not a historical claim that ZFC consciously proceeded through documented Decision Points.

*Remark 6.5* (BOMA and the Set-Theoretic Multiverse). Within the set-theoretic region of the Foundational Space, the Decision Points corresponding to large cardinal axioms [20, 22], forcing-related extensions [8, 9], and variants of the Axiom of Choice [30] generate a cluster of paths whose structure is analogous to Hamkins’ set-theoretic multiverse [16].

The architectural framework adds two elements that Hamkins’ multiverse does not contain. First, the architectural record: the Decision Point at which the large cardinal axiom was injected, the alternatives, and the rationale are documented within the path. Second, the architectural distance: the ultra-pseudometric of Proposition 4.7 provides a measure of how far any two set-theoretic paths are from their common ancestry.

## 6.2 Peano Arithmetic and Its Extensions

*Observation 6.6* (Peano Arithmetic as an Architectural Path). Peano Arithmetic [34] corresponds to a locally consistent path  $\mathcal{P}_{\text{PA}}$  whose Atomic Block declares: a distinguished element 0, a unary successor function  $S$ , and equality; classical first-order

logic; the natural numbers as domain.

*Observation 6.7* (Decision Points in the Peano Development). The most consequential episode is the *Induction Strength Episode*. Options include: first-order induction [34]; second-order induction, yielding  $Z_2$  [39]; and intermediate systems studied in reverse mathematics [11, 39].

*Remark 6.8*. The program of reverse mathematics, developed by Friedman [11] and extended by Simpson [39], is, from the architectural perspective, a systematic investigation of a specific region of the Foundational Space. Its central question — which axioms are necessary and sufficient for specific mathematical theorems? — is an architectural question.

### 6.3 Dependent Type Theory

*Observation 6.9* (Type Theory as an Architectural Path). Martin-Löf Type Theory [26, 27] corresponds to a locally consistent path  $\mathcal{P}_{\text{MLTT}}$  whose Atomic Block declares: types and terms as primitive, with judgment forms  $A$  type,  $a : A$ ,  $A = B$  type, and  $a = b : A$ ; the propositions-as-types interpretation.

*Observation 6.10* (Decision Points Within Type Theory). The *Extensionality Episode* concerns whether the identity type  $\text{Id}_A(a, b)$  [27] is required to satisfy uniqueness of identity proofs [26]. The selection of the non-uniqueness option yields intensional type theory, which is the foundation on which HoTT is built.

The *Universe Hierarchy Episode* concerns whether the type theory includes a cumulative hierarchy  $\mathcal{U}_0 : \mathcal{U}_1 : \dots$ . Different selections produce different type theories with different expressive powers.

### 6.4 Homotopy Type Theory

*Observation 6.11* (HoTT as an Architectural Branch). HoTT [41, 42] corresponds to a locally consistent path branching from  $\mathcal{P}_{\text{MLTT}}$  at the *Univalence Episode*. The Univalence Axiom is not derivable within intensional MLTT and therefore represents a new foundational commitment. Separation formulas for this Decision Point include statements provable in HoTT but not in MLTT.

### 6.5 Categorical Foundations

*Observation 6.12* (Categorical Foundations as Architectural Paths). ETCS, developed by Lawvere [23], corresponds to a locally consistent path  $\mathcal{P}_{\text{ETCS}}$  whose Atomic Block declares: objects, morphisms, composition, and identity as primitive [25, 3]; sets understood as objects of a specific category rather than extensionally determined collections.

*Remark 6.13.* It is known that ETCS and a bounded version of ZFC are mutually interpretable [23, 28]. Within the architectural framework, this mutual interpretability corresponds to a structural relationship between the two paths — a map between their deductive closures preserving logical consequence in both directions. This does not collapse the architectural distinction between the two paths.

## 6.6 The Architectural Status of Existing Foundations

**Proposition 6.14** (Existing Foundations as Architecturally Opaque Paths). *Every existing foundational system corresponds to a locally consistent path in an appropriate Foundational Space. Every such path is valid in the sense of Definition 3.39. No existing foundational system is rendered invalid, superseded, or diminished by BOMA.*

*Proof.* Local consistency of each system follows from its consistency. The remaining claims follow from the definitions: every locally consistent path is a point in the Foundational Space, and BOMA places no restriction on content beyond local consistency and the general structural requirements of Definition 3.25.  $\square$

**Proposition 6.15** (Existing Foundations as Special Cases). *Every existing foundational system, viewed as an architectural path, satisfies the structural requirements of Definition 3.25 but does not satisfy Architectural Transparency as defined in Definition 3.58. Specifically, existing foundations are decision-opaque: their Decision Points are not documented within the formal presentation.*

*Remark 6.16.* Architectural opacity is not a defect; it is a property that follows from the design goals of the existing paradigm. The application of BOMA to existing foundations is an instance of architectural reconstruction in the sense of Remark 6.1: a retrospective interpretive reading. Existing foundations remain mathematically valid, internally consistent, and enormously productive.

## 6.7 Compatibility and Non-Competition

**Principle 6.17** (Non-Competition). BOMA does not compete with any existing foundational system. It does not claim that ZFC is incorrect, that Peano Arithmetic is inadequate, that type theories are misguided, or that categorical foundations are inferior to any alternative.

**Principle 6.18** (Additive Extension). BOMA is an *additive extension* of the existing foundational paradigm: it adds the structural dimension of documented Decision Points, explicit injection structure, and maintained Branch Trees, without removing or altering any content that existing presentations provide.

## 6.8 The Architecture as a Meta-Framework

**Definition 6.19** (Meta-Framework). A *meta-framework* for foundational mathematics is a structure within which individual foundational systems can be situated, compared, and related to one another, without itself constituting a foundational system.

**Proposition 6.20.** *BOMA is a meta-framework in the sense of Definition 6.19 with respect to the object logic of individual paths: it imposes no constraint on which logic any particular path declares. At the level of its own formulation, BOMA employs classical first-order logic as its framework logic — a commitment that is explicit and acknowledged as a meta-level Decision Point in the sense of Remark 3.35.*

*Proof.* The framework makes no specific mathematical commitments at the object level. The additional qualification — that logical neutrality holds at the object level but not at the framework or meta-logic levels — follows from the explicit use of classical reasoning in the framework’s own definitions and proofs, as established in Remark 3.35.  $\square$

*Remark 6.21* (The Meta-Level Decision Point). The choice of classical logic as the framework logic is, by BOMA’s own standards, a Decision Point. The present paper selects classical logic on grounds of familiarity, expressive adequacy, and consistency with the conventions of the mathematical logic community. The unchosen alternative — an intuitionistic formulation of the same architectural framework — is preserved in the record by this acknowledgment.

## 7 Applications and Research Program

### 7.1 Formalization and Proof Assistants

*Observation 7.1* (The Current State of Proof Assistant Foundations). Existing proof assistants are grounded in specific foundational systems: Coq [40] and Agda [33] in variants of MLTT; Lean [31] in a dependent type theory; Isabelle [32] in Higher-Order Logic; Metamath [29] in a metalogical framework. Each of these is, in the architectural sense, an architecturally opaque path.

*Application 7.2* (Architecturally Transparent Proof Assistants). BOMA suggests a new design requirement for proof assistants: not merely verification of formal derivations, but documentation of the architectural choices made in the underlying foundational development, maintained as a first-class structural artifact.

*Application 7.3* (Dependency Analysis and Axiom Tracking). Given an architecturally transparent development, the injection set of every theorem — the collection of

assumptions on which it ultimately depends — is structurally determined by the architectural record. In cases where the underlying logic supports effective proof search, this injection set is mechanically extractable. This enables *axiom profiling*: for any theorem, a precise account of which foundational commitments are necessary for its proof.

A user of a large formal library — such as Mathlib in Lean [24], or the Coq standard library [40] — who wishes to understand why a specific definition was chosen over an alternative currently has no structural source for this information within the library. An architecturally transparent library would make this a first-class structural artifact.

*Application 7.4* (Library Branching and Foundational Experiments). The Branch structure of BOMA suggests a new mode of interaction with formal libraries: the ability to branch a library at a specific Decision Point and explore the consequences of a different foundational choice, while preserving the original development unchanged.

## 7.2 Philosophy of Mathematics

*Application 7.5* (A Formal Framework for Foundational Pluralism). The Foundational Space  $\mathfrak{F}(\beta_0)$  provides a formal framework for foundational pluralism. Within BOMA, foundational pluralism is not a philosophical thesis to be argued for or against; it is a structural fact.

*Application 7.6* (Independence Results as Architectural Information). The independence results of mathematical logic — Gödel’s incompleteness theorems [12], the independence of the Continuum Hypothesis from ZFC [8, 9, 13, 14] — have a precise architectural interpretation. A statement  $\varphi$  independent of a foundation  $\mathcal{P}$  is a separation formula for the Decision Point at which  $\mathcal{P}$  must make a choice about  $\varphi$ . Independence results are architectural certifications of non-trivial Decision Points.

## 7.3 Mathematical Education

*Application 7.7* (Teaching Mathematics Through Decision Points). The architectural discipline suggests presenting mathematical content through its Decision Points: making visible at each stage the choices that were made and the alternatives that existed.

*Application 7.8* (Curriculum Design and Dependency Transparency). An architecturally transparent mathematical development provides a precise dependency structure from which a well-ordered curriculum can be read directly. The partial order on Blocks determines logically admissible orderings of curriculum topics.

## 7.4 Foundations Research

*Application 7.9* (Systematic Mapping of the Foundational Space). The Foundational Space  $\mathfrak{F}(\beta_0)$  is an object of mathematical study in its own right. Systematic mapping of specific regions — for example, the region containing classical and constructive set theories — is a research program that **BOMA** makes coherent and tractable.

*Application 7.10* (Classification of Decision Points). Decision Points vary in the depth and breadth of their mathematical consequences. Their classification by semantic signatures, independence relations, and architectural coupling constitutes a theory of foundational choice.

*Application 7.11* (Minimal Foundations). The Minimal Injection Principle suggests identifying minimal foundations: paths that achieve a specified mathematical result with the smallest possible cumulative injection weight. This extends the reverse mathematics of Friedman [11] and Simpson [39] to the full Foundational Space.

*Application 7.12* (Formal Comparison of Foundational Systems). The structural comparison framework of §5 enables comparison at the level of architectural structure rather than interpretability or consistency strength.

The notion of a Minimal Branch — as defined in Definition 3.34 — is particularly relevant here: a minimal Branch isolates the effect of a single Decision Point selection as cleanly as possible, making the structural comparison maximally informative.

## 7.5 Connections to Adjacent Fields

*Connection 7.13* (Software Engineering and Version Control). The architectural framework is structurally analogous to version-controlled software development [40, 31]. Blocks correspond to commits; dependency structures correspond to commit histories; Decision Points correspond to branch points; Branches correspond to repository branches. The formal infrastructure of version control systems — directed acyclic graphs of commits, branch management — is a computational realization of structures closely related to those defined in **BOMA**.

*Connection 7.14* (Formal Epistemology). The notion of a traceable knowledge system (Definition 4.16) connects **BOMA** to formal epistemology. A traceable knowledge system makes provenance a formally defined structural concept, providing a mathematical model for one of the central concepts of formal epistemology.

*Connection 7.15* (Artificial Intelligence and Mathematical Reasoning). A system trained on architecturally transparent mathematical content would have access to a richer structural representation: not merely the content of mathematical developments, but the architectural choices that shaped them and the alternatives that were available.



## 7.6 The Immediate Research Agenda

**Formal Verification of the Framework.** The encoding of BOMA’s components in a proof assistant and the mechanically verified proof of its main propositions. This would produce the first architecturally transparent formal development — a development of BOMA itself within the architectural paradigm.

**Reconstruction of a Classic Foundation.** The reconstruction of ZFC or Peano Arithmetic as an architecturally transparent path, producing the first systematic Decision Point Catalogue for a major foundational system.

**Development of Comparison Tools.** Algorithms for computing common ancestries, divergence points, semantic signatures, and injection divergences from structural records.

**First Branch Developments.** The systematic exploration of what follows from alternative selections at specific, architecturally characterized Decision Points — beginning with the Decision Points identified in §6.1 through §6.5.

## 8 Conclusion

### 8.1 What This Paper Has Not Done

This paper has not proposed a new mathematical foundation. It has not introduced new axioms, defined new primitive notions, or derived new mathematical theorems. It has not claimed that existing foundations are deficient. It has not resolved the philosophical questions surrounding foundational pluralism. It has not developed any Branch in full. The architecture has been defined; its application is the work of subsequent research.

### 8.2 What This Paper Has Done

We have identified an architectural gap in the existing foundational paradigm: not a logical gap, but an organizational one, concerning the documentation of Decision Points, the preservation of alternatives, and the declaration of injection structure.

We have proposed BOMA, a precise architectural framework addressing this gap through seven structural components.

We have introduced the Foundational Space as the global object produced by the architectural framework, established inter-path contradictions as structural information (Theorem 4.14), and characterized architecturally transparent paths as traceable knowledge systems (Proposition 4.17).



We have established comparison as an epistemological tool (Theorem 5.7) and demonstrated that BOMA is a meta-framework within which existing foundations — ZFC [20, 44], Peano Arithmetic [34], Martin-Löf Type Theory [26, 27], HoTT [41], and categorical foundations [23, 25] — are situated as locally consistent, architecturally opaque paths.

### 8.3 The Central Claim

The central claim of this paper is the following.

The organizational paradigm of existing foundational mathematics — linear presentation of axioms, definitions, and theorems, with Decision Points invisible, alternatives suppressed, and injection structure undeclared — is one paradigm among possible paradigms [5, 37], and not the only one consistent with mathematical rigor.

BOMA is an alternative paradigm: one in which every foundational development is organized as a Branch Tree of locally consistent paths, every genuine choice is recorded at the moment it is made, every new assumption is individually declared, and every development is situated within a navigable space of alternative developments.

### 8.4 What Comes Next

The immediate next steps are those identified in §7.6: formal verification, reconstruction of a classic foundation, development of comparison tools, and first Branch developments.

*Remark 8.1* (Scope and the Broader Vision). The present paper has demonstrated BOMA at the level of foundational systems. This is the appropriate starting point, but not the intended limit of the framework’s scope.

The architectural discipline of BOMA applies at every level of mathematical development. A definition in analysis, a lemma in algebra, a construction in topology — each is a Block in the sense of Definition 3.1. A researcher who wishes to explore two competing formulations of a definition does not face a foundational question; they face a Decision Point at the level of their immediate mathematical development. BOMA applies there with full generality.

This scope extension carries a methodological consequence that the present paper has not foregrounded: the architectural discipline makes mathematical development experimentally tractable. Trial and error — the exploration of alternatives through deliberate branching, observation of consequences, and revision of choices — becomes a first-class mathematical methodology rather than an informal practice. The Branch Tree is the formal record of this experimental process; local consistency is the criterion that makes each experimental path self-validating.

The full realization of this vision — mathematics as a constructible, branchable, experimentally navigable structure at every level of development, including levels not yet reached by existing mathematics — is the long-term research program of which this paper is the architectural foundation.

Within the Foundational Space  $\mathfrak{F}(\beta_0)$ , unknown mathematics occupies a determinate structural position: the collection of locally consistent paths that have not yet been traversed. These paths are reachable from existing developments through finite sequences of Decision Point reselections (Definition 4.12). The act of mathematical exploration, within the architectural paradigm, is the act of extending a Branch into an untraversed region of  $\mathfrak{F}(\beta_0)$ . Mathematics is inexhaustible not because its objects are infinite in number, but because the Foundational Space is infinite in structure. BOMA does not close this space; it makes its infinitude navigable.

## 8.5 A Remark on Method

This paper has been written in a style of deliberate tentativeness. It claims what it can support and declines to claim what it cannot. It characterizes existing foundations with care, attributing to them the substantial achievements they have and declining to attribute failures they do not have.

This tentativeness is not diffidence. It is an expression of the Minimal Injection Principle applied to the domain of intellectual claims: a paper should claim no more than is supported by its arguments, and present no derived conclusion as an injected assumption. The architecture is offered as the minimum structure needed to make the architectural dimension of foundational mathematics visible and tractable.

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